Design and Analysis of Interferometric Structures for Group Index Extraction in Silicon Photonics

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***Abstract*—This work presents the design, simulation, fabrication, and experimental testing of integrated interferometric photonic circuits for the extraction of the group index. Two different Mach–Zehnder interferometer (MZI) architectures are investigated. The first design is based on conventional strip waveguides in both interferometer arms, while the second design includes a grating waveguide in one arm, enabling the analysis of slow-light effects. In both cases, the interferometers are intentionally unbalanced to allow the extraction of the group index from the measured free spectral range (FSR).**

***Index Terms*—Integrated photonics, Mach-Zehnder interferometer, group index extraction, free spectral range, slow-light waveguides, sub-wavelength gratings.**

I. Introduction

Silicon photonics has emerged as a key technology for the realization of compact, low-cost, and high-performance optical integrated circuits, enabling applications in optical communications, sensing, and signal processing. Silicon photonics leverages the existing CMOS manufacturing infrastructure, enabling large-scale production [1]. Among integrated photonic devices, interferometric circuits such as the Mach-Zehnder interferometer play a central role in modulation, filtering, and sensing applications. In particular, unbalanced interferometers provide a simple and robust method to extract the group index by measuring the free spectral range of the transmission spectrum [1], [2]. The objective of this project is the design, simulation, fabrication, and experimental characterization of integrated MZI circuits for the extraction of the group index. Two different interferometer architectures are investigated: a conventional strip-waveguide MZI and a MZI incorporating a grating waveguide to analyze slow-light effects.

II. Theoretical Analyses

A. Waveguide Compact Model

The optical properties of a waveguide can be described using a compact model based on the effective refractive index $n_{\text{eff}}(\lambda)$ and group index. The propagation constant $\beta(\lambda)$ is given by

$$\beta(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}}(\lambda). \tag{1}$$

The group index is defined as

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}(\lambda)}{d\lambda}. \tag{2}$$

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B. Mach–Zehnder Interferometer Transfer Function

An unbalanced MZI consists of two arms with a path length difference $\Delta L = L_2 - L_1$. Assuming ideal 50/50 splitters, the normalized transmission at the output port can be expressed as

$$T_{\text{MZI}}(\lambda) = \frac{1}{4} \left| e^{-i\beta L_1} + e^{-i\beta L_2} \right|^2 \tag{3}$$

The free spectral range (FSR) of the interferometer is related to the group index by

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g(\lambda)\Delta L}. \tag{4}$$

III. Modelling and Simulation

A. Waveguide Geometries and Circuit Designs

Two interferometric circuits are considered, each implemented in two distinct configurations. Both interferometric circuits are designed for quasi-TE polarization using strip waveguides with a fixed silicon thickness of $220\,\text{nm}$, as imposed by the fabrication process and a waveguide width of $500\,\text{nm}$. In the second interferometric architecture, one arm incorporates a grating waveguide. The grating is characterized by an outer width of 750 nm, an inner width of 250 nm, and a grating period of $0.305\,\mu\text{m}$. These parameters are chosen to induce slow-light effect and to position the band edge of the dispersion diagram around $1543\,\text{nm}$. Two grating lengths are investigated, corresponding to 850 and 1200 periods. In addition to the interferometric devices, two reference structures consisting of grating waveguides placed between two grating couplers, with the same geometrical parameters as those employed in the two MZI configurations, are included to enable an independent characterization of the two gratings. Finally, another reference structure is included, consisting of two grating couplers connected by a strip waveguide. For circuit interconnections and power splitting, standard components from the EBeam library are employed, including the broadband directional coupler `ebeam_bdc_te1550` and the adiabatic Y-branch splitter `ebeam_y_adiabatic_500pin`.

B. Strip Waveguide Mode Analysis

The fundamental guided mode of the strip waveguide is simulated using *Lumerical MODE Solutions*. The simulated electric field amplitude profile of the fundamental quasi-TE mode is shown in Fig. 1. This mode exhibits an effective index of $n_{\text{eff}} = 2.44$ at a wavelength of $1550\,\text{nm}$, and a corresponding group index of $n_g = 4.20$.

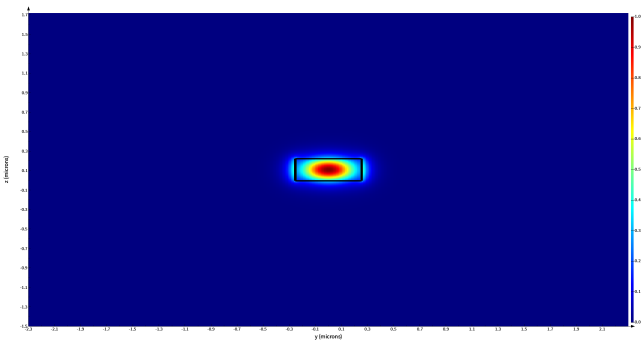

Fig. 1. Simulated fundamental quasi-TE mode profile for a strip waveguide with width $500\,\mathrm{nm}$ and height $220\,\mathrm{nm}$.

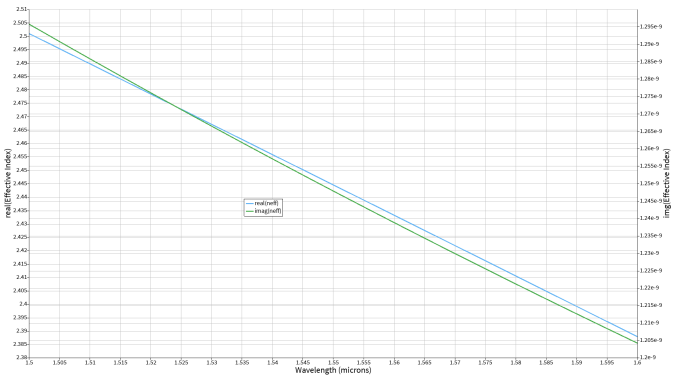

Fig. 2. Simulated effective index n_{eff} as a function of wavelength for the strip waveguide.

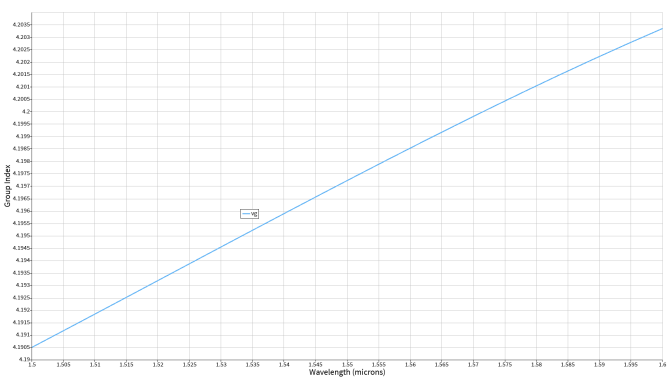

Fig. 3. Simulated group index n_g as a function of wavelength for the strip waveguide.

C. Effective and Group Index Simulations

The effective index n_{eff} and group index n_g of the strip waveguide are computed as a function of wavelength over the range $1500\,\mathrm{nm}$ to $1600\,\mathrm{nm}$. These are obtained directly from the finite difference eigenmode simulations in Lumerical Mode. Figure 2 reports the simulated effective index as a function of wavelength, while Fig. 3 shows the group index.

D. Waveguide Compact Model (Strip, $width = 500\,\mathrm{nm}$, TE)

A compact model for the strip waveguide is derived from the simulated effective index data exported from *Lumerical MODE Solutions*. The dataset consists of 20 wavelength points uniformly distributed in the $1.50\,\mu\mathrm{m}$ to $1.60\,\mu\mathrm{m}$ range, together with the corresponding complex effective index values $n_{\mathrm{eff}}(\lambda)$. A second-order Taylor expansion around the center wavelength $\lambda_0 = 1.55\,\mu\mathrm{m}$ is adopted as the waveguide compact model:

$$n_{\mathrm{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2. \tag{5}$$

The polynomial coefficients are estimated using a fitting procedure implemented in MATLAB. For the strip waveguide with width $500\,\mathrm{nm}$ and quasi-TE polarization, the fitted parameters are:

$$[a_0,\ a_1,\ a_2] = [2.44,\ -1.13,\ -0.04]\,. \tag{6}$$

The fitted model shows excellent agreement with the simulated data over the entire wavelength range, confirming the validity of the second-order approximation. From the compact model, the group index is obtained as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}, \tag{7}$$

where

$$\frac{dn_{\mathrm{eff}}}{d\lambda} = a_1 + 2a_2(\lambda - \lambda_0). \tag{8}$$

E. Adiabatic Y-branch Design and Characterization

The optical power splitting in the interferometric circuits is implemented using an adiabatic Y-branch in order to ensure low reflection, broadband operation, and equal power splitting over the wavelength range of interest. The Y-branch is first analyzed using a variational FDTD approach to obtain a preliminary estimate of the transmission spectrum and to verify correct power splitting behavior. Subsequently, a full 3D FDTD simulation is performed to extract the scattering parameters (S-parameters) of the device. A mode source exciting the fundamental quasi-TE mode is placed at the input waveguide (Port 1). Three frequency-domain monitors are positioned at the input and output ports to record the transmitted and reflected power. The wavelength sweep spans the range $1.50\,\mu\mathrm{m}$ to $1.60\,\mu\mathrm{m}$. The simulated transmission spectra at the three ports are shown in Fig. 4, Fig. 5 and Fig. 6. When light is injected from Port 1, the optical power is evenly split between the two output ports, with each arm carrying approximately 50% of the input power across the entire wavelength range.

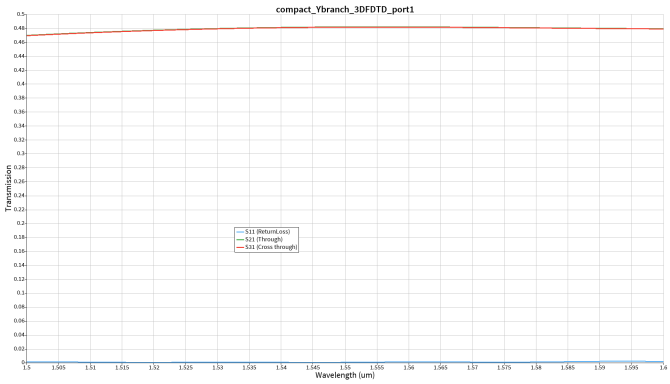

Fig. 4. Simulated transmission and reflection coefficients of the adiabatic Y-branch obtained from 3D FDTD simulations when the fundamental TE mode is injected into Port 1.

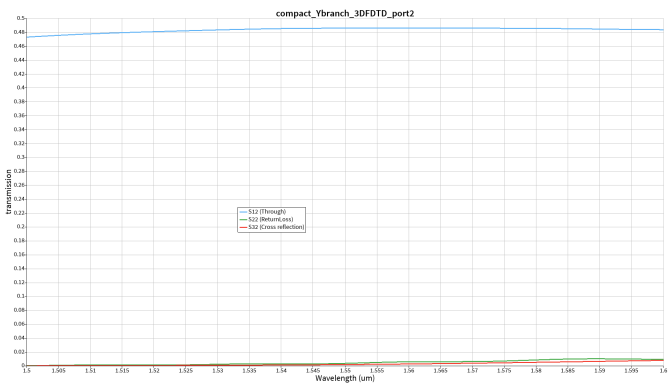

Fig. 5. Simulated transmission and reflection coefficients of the adiabatic Y-branch obtained from 3D FDTD simulations when the fundamental TE mode is injected into Port 2.

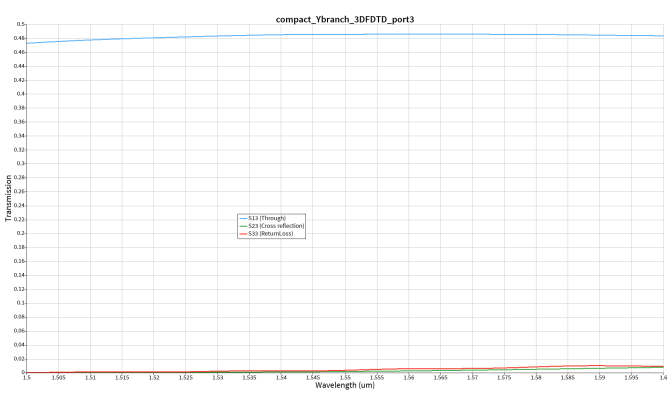

Fig. 6. Simulated transmission and reflection coefficients of the adiabatic Y-branch obtained from 3D FDTD simulations when the fundamental TE mode is injected into Port 3.

F. Mach–Zehnder Interferometer Transfer Function

The optical transfer function of an unbalanced Mach-Zehnder interferometer (MZI) can be expressed as a function of the optical wavelength by considering the phase difference accumulated between the two interferometer arms. Assuming ideal $50:50$ couplers and equal propagation losses in both arms, the normalized transmission at the output port of an imbalanced interferometer with identical waveguides can be expressed as

$$T(\lambda) = \frac{1}{2}\left[1 + \cos(\Delta\phi(\lambda))\right], \tag{9}$$

where $\Delta\phi(\lambda)$ is the wavelength-dependent phase difference between the two arms. The phase difference is given by

$$\Delta\phi(\lambda) = \frac{2\pi}{\lambda}\, n_{eff}(\lambda)\, \Delta L, \tag{10}$$

where $\Delta L = L_2 - L_1$ is the path length difference between the two interferometer arms.
The periodicity of the transfer function in wavelength corresponds to the free spectral range (FSR), defined as

$$\mathrm{FSR}(\lambda) = \frac{\lambda^2}{n_g(\lambda)\, \Delta L}. \tag{11}$$

G. Parameter variations and expected performance (FSR check)

Two identical unbalanced MZI modulators were implemented using quasi-TE strip waveguides ($220\,\mathrm{nm}$ Si thickness, $500\,\mathrm{nm}$ width). The two layouts differ only in the path-length imbalance $\Delta L = L_2 - L_1$.
The selected path length differences ΔL result in free spectral ranges of $2.71\,\mathrm{nm}$ for MZI_1 ($\Delta L = 210.10\,\mu\mathrm{m}$) and $1.47\,\mathrm{nm}$ for MZI_2 ($\Delta L = 397.10\,\mu\mathrm{m}$) at $\lambda = 1550\,\mathrm{nm}$. Over the wavelength range $1500\,\mathrm{nm}$ to $1600\,\mathrm{nm}$, the FSR remains almost constant (Fig. 7). These FSR values are compatible with laboratory measurements and are summarized in Table I. Using the simulated free spectral range (FSR) at $1550\,\mathrm{nm}$ and applying equation (11), the group index was extracted, yielding $n_g = 4.22$ for $\Delta L = 210.10\,\mu\mathrm{m}$ and $n_g = 4.12$ for $\Delta L = 397.10\,\mu\mathrm{m}$.

TABLE I
EXPECTED PERFORMANCE FOR THE TWO MZI

Modulator	*width* (nm)	ΔL (µm)	FSR@1550 nm (nm)	n_g
MZI_1 (smaller ΔL)	500	210.10	2.71	4.22
MZI_2 (larger ΔL)	500	397.10	1.47	4.12

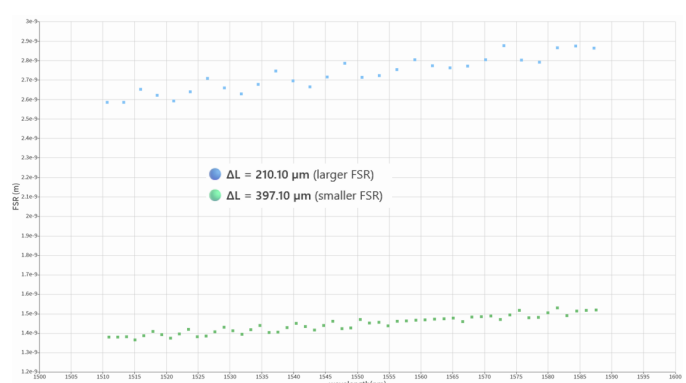

Fig. 7. Simulated free spectral range (FSR) as a function of wavelength for the two MZI modulators with different path-length imbalance ΔL.

H. Transmission Spectrum of the Mach–Zehnder Interferometers

Figure 8 shows the simulated transmission spectra of the two Mach–Zehnder interferometers, obtained from circuit-level simulations. The spectra are reported as optical gain versus wavelength for the two devices, which differ only in the path length difference ΔL.

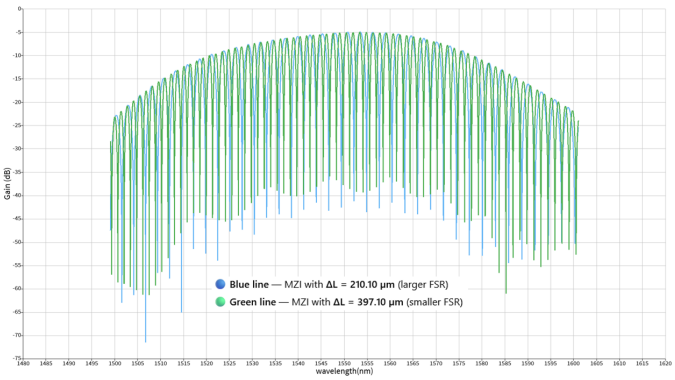

Fig. 8. Simulated transmission spectra of the two Mach–Zehnder interferometers with different path length ΔL.

In both cases, the transmission spectrum is modulated by a smooth envelope centered around $1550\,\mathrm{nm}$, mainly determined by the grating couplers.

I. Effect of Slow-Light Grating

An analogous analysis is carried out for the additional two Mach–Zehnder interferometers (MZI_3 and MZI_4), which differ from MZI_1 and MZI_2 by the presence of a slow-light grating waveguide in one interferometer arm. For MZI_3 and MZI_4, the corrugation width and periods are identical, while the grating lengths are $1200\,\mu\mathrm{m}$ and $850\,\mu\mathrm{m}$, respectively. For each device, the length of the reference arm (without grating) is adjusted such that it matches the length of the grating arm excluding the grating section. Figure 9 shows the simulated transmission spectra of MZI_3 and MZI_4. Close to the band edge, the transmission spectra become irregular. This effect arises from a strong backscattering.

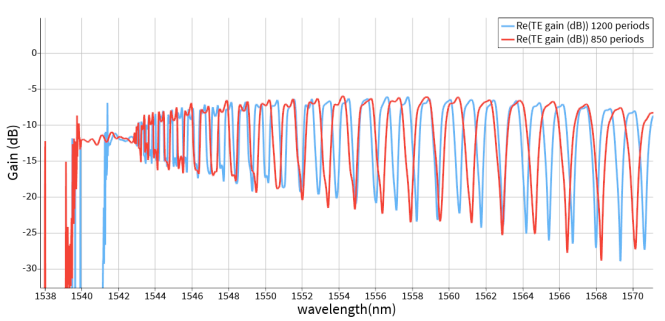

Fig. 9. Simulated transmission spectra of MZI_3 and MZI_4, including a slow-light grating waveguide with 1200 and 850 periods, respectively.

The corresponding free spectral range extracted from the transmission spectra is reported in Fig. 10. In contrast to the nearly constant FSR observed for MZI_1 and MZI_2, the FSR of MZI_3 and MZI_4 decreases significantly as the wavelength approaches the band edge. This behavior is a direct consequence of the strong increase of the group index near the grating band edge in the slow-light regime.

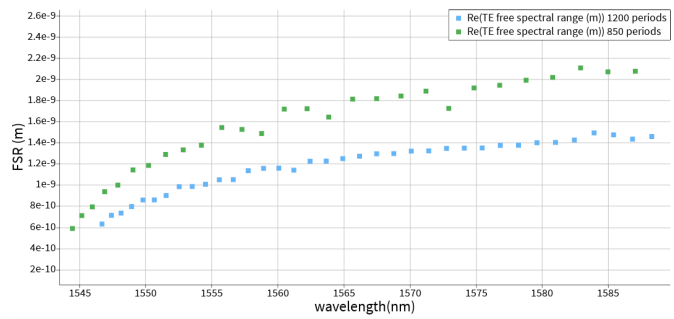

Fig. 10. Free spectral range as a function of wavelength for MZI_3 and MZI_4. The FSR decreases as the wavelength approaches the grating band edge due to the increasing group index.

IV. Fabrication

The photonic devices presented in this work were fabricated thanks to edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course. Fabrication was performed through MPW processes at Applied Nanotools Inc. and the Washington Nanofabrication Facility (WNF), using silicon-on-insulator (SOI) technology and electron-beam lithography technique [4]. The devices were characterized using a standard automated optical measurement setup [5]–[7]. Figure 11 shows the layout of the chip.

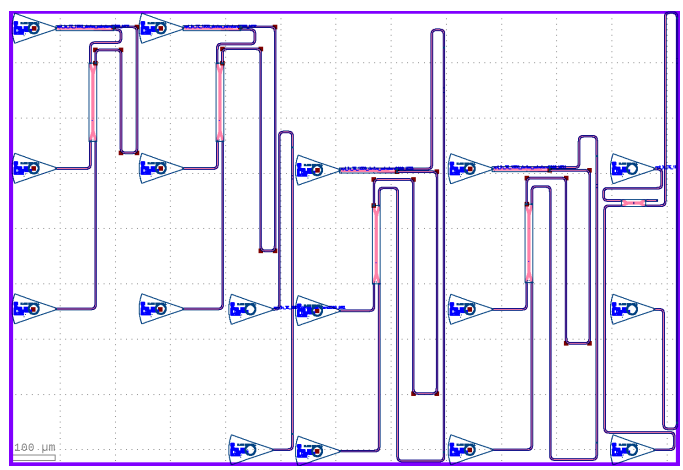

Fig. 11. Chip layout used for fabrication.

V. Manufacturing Variability and Corner Analysis (Strip Waveguide)

In this section, the impact of fabrication tolerances on the performance of the strip waveguide with nominal width of $500\,\mathrm{nm}$ is analyzed. Process variations in both waveguide width and thickness are considered, and a corner analysis is performed to evaluate their effect on the effective index n_{eff}, group index n_g, and the resulting free spectral range

(FSR) of the interferometric circuits. The considered parameter space includes width variations from $470\,\mathrm{nm}$ to $530\,\mathrm{nm}$ and thickness variations from $215.3\,\mathrm{nm}$ to $223.1\,\mathrm{nm}$.

TABLE II
CORNER ANALYSIS RESULTS FOR STRIP WAVEGUIDES.

Width	Thickness	n_{eff}	n_g	FSR ($\Delta L = 210.10\mu$m)	FSR ($\Delta L = 397$
470	215.3	2.3703	4.229	2.704	1.431
470	223.1	2.3987	4.237	2.699	1.428
490	215.3	2.4050	4.191	2.729	1.444
490	223.1	2.4335	4.198	2.724	1.441
500	220.0	2.4404	4.177	2.738	1.448
510	215.3	2.4380	4.158	2.750	1.455
510	223.1	2.4661	4.165	2.746	1.453
530	215.3	2.4655	4.126	2.771	1.466
530	223.1	2.4936	4.132	2.767	1.464

From the corner analysis, the following variations are observed:

- Minimum n_g = 4.126
- Maximum n_g = 4.237
- Minimum FSR ($\Delta L = 210.10\,\mu\mathrm{m}$) $= 2.699nm$
- Maximum FSR ($\Delta L = 210.10\,\mu\mathrm{m}$) $= 2.771nm$
- Minimum FSR ($\Delta L = 397.10\,\mu\mathrm{m}$) $= 1.428nm$
- Maximum FSR ($\Delta L = 397.10\,\mu\mathrm{m}$) $= 1.466nm$

At fixed thickness, increasing the waveguide width leads to an increase in the effective index n_{eff}, a decrease in the group index n_g, and a corresponding increase in the FSR.
At fixed width, increasing the waveguide thickness results in an increase in both the effective index n_{eff} and the group index n_g, and therefore a reduction in the FSR.

VI. EXPERIMENTAL ANALYSIS

A. Extraction Method for n_g and FSR

The extraction of the group index n_g and the free spectral range (FSR) from the experimental data was performed using a MATLAB code provided by the course. First, the measured transmission spectrum is baseline-corrected by fitting a low-order polynomial and subtracting it from the raw data. The corrected spectrum is then smoothed and the transmission minima are identified using a peak-finding algorithm. From the wavelength spacing between consecutive minima, the free spectral range is calculated as

$$\mathrm{FSR} = \lambda_{i+1} - \lambda_i. \tag{12}$$

The group index is then obtained from the relation

$$n_g(\lambda) = \frac{\lambda^2}{\mathrm{FSR} \cdot \Delta L}. \tag{13}$$

These values are used as initial estimates for a global fitting of the Mach–Zehnder interferometer (MZI) transfer function. The final values of n_g is extracted from the fitted model.

B. MZI$_1$ Experimental vs Simulated Results

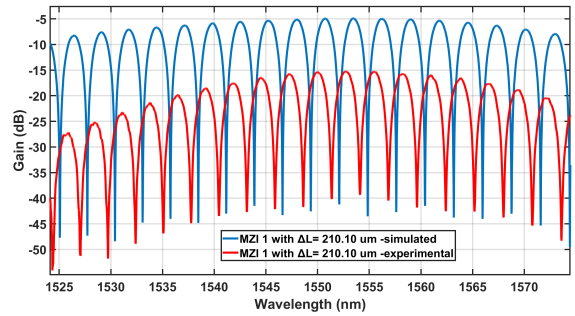

Fig. 12. Comparison between simulated and experimental transmission spectra for MZI$_1$ ($\Delta L = 210.10\,\mu\mathrm{m}$).

Figure 12 shows the comparison between the simulated and experimental transmission spectra for MZI$_1$. The experimental data have been obtained using the IRIS dataset. The discrepancy in the extinction ratio between simulated and experimental curves is mainly attributed to losses introduced by fiber coupling and photodetectors. Moreover, the lack of knowledge of the input optical power in the IRIS dataset results in an unknown vertical offset.

At a wavelength of $1550\,\mathrm{nm}$, the extracted parameters are:

- FSR (experimental): $2.738\,\mathrm{nm}$
- FSR (simulated): $2.743\,\mathrm{nm}$
- n_g (experimental): 4.176
- n_g (simulated): 4.169

The relative discrepancy on the group index is

$$\Delta n_g \approx \frac{|4.176 - 4.169|}{4.169} \times 100 \approx 0.17\%. \tag{14}$$

This very small deviation confirms an excellent agreement between simulation and experiment. Furthermore, the experimental values lie within the interval predicted by the corner analysis reported in Table II.

C. MZI$_2$ Experimental vs Simulated Results

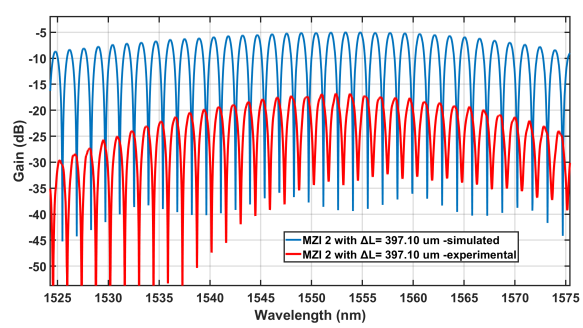

Fig. 13. Comparison between simulated and experimental transmission spectra for MZI$_2$ ($\Delta L = 397.10\,\mu\mathrm{m}$).

Figure 13 shows the comparison between simulated and experimental transmission spectra for MZI$_2$, also obtained from the IRIS dataset. As for MZI$_1$, a discrepancy is also observed for MZI$_2$, due to the same problems.

At $1550\,\mathrm{nm}$, the extracted parameters are:

- FSR (experimental): $1.446\,\mathrm{nm}$
- FSR (simulated): $1.448\,\mathrm{nm}$
- n_g (experimental): 4.185
- n_g (simulated): 4.179

The relative discrepancy on the group index is

$$\Delta n_g \approx \frac{|4.185 - 4.179|}{4.179} \times 100 \approx 0.14\%. \tag{15}$$

Also in this case, the agreement between simulation and experiment is excellent. Also in this case, the experimental results fall within the bounds predicted by the corner analysis.

D. MZI$_3$ (Long Grating, $\Delta L = 366$ μm) Experimental vs Simulated Results

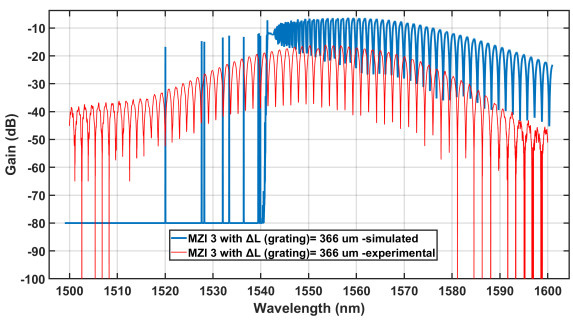

Fig. 14. Simulated and experimental spectra of MZI$_3$ with $\Delta L = 366$ μm.

Figure 14 shows a significant discrepancy between simulated and experimental results. The bandgap clearly visible in simulations is not observed in the experimental data. As a result, the slow-light effect is not present in the measurements, and the two curves are not directly comparable. The grating period was finding using Lumerical interconnect, selecting $a = 305$ nm to place the band edge around ~ 1543 nm. However, MPB photonics bands simulations of the same structure predict the band edge at ~ 1455.76 nm. Fabrication variability must also be considered, as it can cause a shift of the spectrum. The same behavior is observed for MZI$_4$ ($\Delta Lgrating = 259.25$ μm), leading to a similar mismatch between simulations and measurements.

VII. Acknowledgements

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Enxiao Luan performed the measurements at The University of British Columbia. I acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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